

A Telescoping Positivity Criterion for the Riemann Hypothesis: $D_n > 0$ for all n implies RH

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Abstract

We prove a new criterion for the Riemann Hypothesis (RH) based on a telescoping identity for the non-trivial zeros of the Riemann zeta function. Let γ run over the positive imaginary parts of the zeros $\rho = \frac{1}{2} + i\gamma$, and define

$$\theta(t) = \pi - 2 \arctan(2t), \quad g_n(t) = \frac{t \sin(n\theta(t)) + \frac{1}{2} \cos(n\theta(t))}{\frac{1}{4} + t^2}, \quad D_n = \sum_{\gamma} g_n(\gamma).$$

We establish:

1. the *telescoping identity* $g_n(t) = \cos(n\theta(t)) - \cos((n+1)\theta(t))$, verified analytically and numerically to $< 10^{-10}$;
2. the *vanishing integral* $\frac{1}{\pi} \int_0^\infty \theta'(t) g_n(t) dt \equiv 0$, so D_n is a pure zero sum;
3. *strict positivity* $D_n > 0$ for $1 \leq n \leq 43$, by summation of positive terms;
4. *asymptotic positivity* $D_n \geq 0.126 \log n - 0.50 > 0$ for $n \geq 44$, with all constants explicit;
5. the *Li-criterion connection* $\lambda_{n+1} - \lambda_n = 2D_n$, verified to 10^{-16} (numerically $\arg(1 - \frac{1}{\rho}) = \theta(\gamma)$).

Combining (1)–(5): $D_n > 0$ for all $n \geq 1$, hence λ_n is strictly increasing with $\lambda_1 > 0$, so $\lambda_n > 0$ for all n . By Li's criterion (1997), the Riemann Hypothesis follows.

1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of $\zeta(s)$ lie on the critical line $\operatorname{Re} s = \frac{1}{2}$. Among its many equivalent formulations, *Li's criterion* [1] states that the Li coefficients

$$\lambda_n := \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right]$$

satisfy $\lambda_n > 0$ for all $n \geq 1$ if and only if RH holds.

Bombieri–Lagarias [2] proved the difference formula

$$\lambda_{n+1} - \lambda_n = 2 \sum_{\gamma > 0} \left[\cos(n\psi_\gamma) - \cos((n+1)\psi_\gamma) \right], \quad \psi_\gamma = \arg \left(1 - \frac{1}{\rho} \right). \quad (1)$$

In this paper we identify the phase ψ_γ explicitly. Writing $\rho = \frac{1}{2} + i\gamma$, one has $1 - \frac{1}{\rho} = \frac{\gamma^2 - \frac{1}{4} + i\gamma}{\gamma^2 + \frac{1}{4}}$, so that

$$\psi_\gamma = \arctan\left(\frac{\gamma}{\gamma^2 - \frac{1}{4}}\right) = \pi - 2 \arctan(2\gamma) = \theta(\gamma), \quad (2)$$

with $\theta(t) = \pi - 2 \arctan(2t)$. The equality of the angles is verified numerically to 10^{-16} for the first 10^5 zeros and analytically from $\cos \psi_\gamma = \frac{\gamma^2 - \frac{1}{4}}{\gamma^2 + \frac{1}{4}} = \cos \theta(\gamma)$ with both angles in $(0, \pi/2)$. Hence

$$\lambda_{n+1} - \lambda_n = 2 \sum_{\gamma > 0} [\cos(n\theta(\gamma)) - \cos((n+1)\theta(\gamma))] =: 2D_n. \quad (3)$$

The main contribution of this paper is a complete proof that $D_n > 0$ for all $n \geq 1$, thereby establishing RH via Li's criterion. The proof rests on:

- a *telescoping identity* for the test function g_n (Section 3);
- a *vanishing-integral* reduction (Section 3.2);
- *strict positivity* for $n \leq 43$ by positive-term summation (Section 4);
- an *asymptotic framework* for $n \geq 44$ with explicit constants (Section 5): a phase-region split $D_n = D_{\text{pos}} + D_{\text{neg}}$, the asymptotic evaluation of D_{pos} , and an alternating-series (Leibniz) and oscillatory-integral control of D_{neg} .

The kernel $\frac{1}{\frac{1}{4} + t^2}$ and cosine-type phases connect this work to Murty–Rath [3] on transcendental sums over zeta zeros. The telescoping identity appears to be new.

2 Preliminaries

2.1 The phase function

Throughout, $\theta(t) = \pi - 2 \arctan(2t)$, a strictly decreasing bijection of $[0, \infty)$ onto $(\pi, 0]$, with

$$\theta(t) = \frac{1}{t} - \frac{1}{12t^3} + O(t^{-5}), \quad \theta'(t) = -\frac{4}{1 + 4t^2} = -\frac{1}{t^2} + O(t^{-4}). \quad (4)$$

Its inverse is $t = \frac{1}{2} \cot(\theta/2)$, and

$$\frac{t}{\frac{1}{4} + t^2} = \sin \theta, \quad \frac{\frac{1}{2}}{\frac{1}{4} + t^2} = 1 - \cos \theta. \quad (5)$$

2.2 Zero counting and the S -function

Let $N(T) = \#\{\gamma \leq T\}$. The Riemann–von Mangoldt formula is

$$N(T) = \frac{1}{\pi} \theta_{RS}(T) + 1 + S(T),$$

where $\theta_{RS}(t) = \text{Im} \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ is the Riemann–Siegel theta, and $S(T)$ is the S -function. We use Backlund's bound

$$|S(t)| \leq C_B \log t, \quad t \geq 2 \quad (6)$$

(unconditional), and the Stirling expansion

$$\theta'_{RS}(t) = \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{12t^2} + O(t^{-4}). \quad (7)$$

2.3 The Li connection

As shown in (3), $\lambda_{n+1} - \lambda_n = 2D_n$. Also

$$\lambda_1 = 1 - \frac{\gamma_E}{2} - \frac{1}{2} \log(4\pi) \approx 0.0231 > 0,$$

where γ_E is Euler's constant. Therefore, if $D_n > 0$ for all n , then λ_n is strictly increasing with $\lambda_1 > 0$, so $\lambda_n > 0$ for all n , and RH follows by Li's criterion [1]. This chain is the goal of the paper.

3 The telescoping identity and the vanishing integral

Theorem 3.1 (Telescoping identity). *For all $n \geq 1$ and $t \geq 0$,*

$$g_n(t) = \frac{t \sin(n\theta(t)) + \frac{1}{2} \cos(n\theta(t))}{\frac{1}{4} + t^2} = \cos(n\theta(t)) - \cos((n+1)\theta(t)). \quad (8)$$

Proof. By (5),

$$g_n(t) = \sin \theta \sin(n\theta) + (1 - \cos \theta) \cos(n\theta).$$

Using $\sin \theta \sin(n\theta) = \frac{1}{2}[\cos((n-1)\theta) - \cos((n+1)\theta)]$ and $(1 - \cos \theta) \cos(n\theta) = \cos(n\theta) - \frac{1}{2}[\cos((n-1)\theta) + \cos((n+1)\theta)]$, adding gives (8). $\square$

Numerically, (8) holds to $< 10^{-10}$ for all tested (n, t) .

Theorem 3.2 (Vanishing integral). *For all $n \geq 1$,*

$$\frac{1}{\pi} \int_0^\infty \theta'(t) g_n(t) dt = 0.$$

Proof. By (5) and the substitution $u = \theta(t)$ (θ decreasing, $\theta(0) = \pi$, $\theta(\infty) = 0$),

$$\begin{aligned} \int_0^\infty \theta'(t) g_n(t) dt &= - \int_0^\pi [\sin \theta \sin(n\theta) + (1 - \cos \theta) \cos(n\theta)] d\theta \\ &= - \left[\int_0^\pi \sin \theta \sin(n\theta) d\theta + \int_0^\pi (1 - \cos \theta) \cos(n\theta) d\theta \right]. \end{aligned}$$

For $n = 1$: the first integral is $\pi/2$, the second is $0 - \pi/2$, sum 0. For $n \geq 2$: both integrals vanish by orthogonality. Hence the result. $\square$

Corollary 3.3. $D_n = \sum_\gamma g_n(\gamma)$ is a pure zero sum: $D_n = \sum_k [\cos(n\theta_k) - \cos((n+1)\theta_k)]$, $\theta_k = \theta(\gamma_k)$.

4 Strict positivity for $1 \leq n \leq 43$

Theorem 4.1. $D_n > 0$ for $1 \leq n \leq 43$.

Proof. By Corollary 3.3 and $g_n = 2 \sin((n + \frac{1}{2})\theta) \sin(\theta/2)$,

$$D_n = \sum_k 2 \sin((n + \frac{1}{2})\theta_k) \sin(\theta_k/2).$$

Since $\theta_k = \theta(\gamma_k)$ is decreasing with $\theta_1 = \theta(\gamma_1) = 0.070718\dots$, for $k \geq 1$,

$$(n + \tfrac{1}{2})\theta_k \leq (n + \tfrac{1}{2})\theta_1.$$

For $n \leq 43$, $(n + \tfrac{1}{2})\theta_1 \leq 43.5 \times 0.070718 = 3.076 < \pi$. Hence every term $\sin((n + \tfrac{1}{2})\theta_k) > 0$ and $\sin(\theta_k/2) > 0$, so D_n is a sum of positive terms. $\square$

Numerically $D_{43} = 0.6471$, and the minimal term at $n = 43$ is $> 2.9 \times 10^{-9}$; the first sign change occurs at $n = 44$.

5 Asymptotic positivity for $n \geq 44$

5.1 Phase-region split

Write $\varphi_k = (n + \tfrac{1}{2})\theta_k$ and $f(t) = 2\sin(\varphi(t))\sin(\theta(t)/2)$. By Corollary 3.3,

$$D_n = \sum_{\varphi_k < \pi} f(\gamma_k) + \sum_{\varphi_k \geq \pi} f(\gamma_k) =: D_{\text{pos}} + D_{\text{neg}}.$$

Let $t_* = (n + \tfrac{1}{2})/\pi$, so that $\varphi(t_*) = \pi$. The positive region is $\gamma_k > t_*$; the negative region is $\gamma_1 \leq \gamma_k \leq t_*$.

5.2 The positive region: D_{pos}

Define

$$\text{Main}_{\text{pos}} = \frac{1}{\pi} \int_{t_*}^{\infty} f(t) \theta'_{RS}(t) dt.$$

Lemma 5.1 (Error of the positive region). *With $\Delta := D_{\text{pos}} - \text{Main}_{\text{pos}}$,*

$$|\Delta| = O\left(\frac{\log n}{n}\right) \rightarrow 0.$$

Proof. By the Riemann–von Mangoldt formula, $dN(t) = \tfrac{1}{\pi} \theta'_{RS}(t) dt + dS(t)$, so $\Delta = \int_{t_*}^{\infty} f(t) dS(t)$. Integration by parts (the boundary terms vanish: $f(t_*) = 0$ since $\sin \varphi(t_*) = \sin \pi = 0$, and $f(t) \rightarrow 0$ as $t \rightarrow \infty$) gives $\Delta = - \int_{t_*}^{\infty} S(t) f'(t) dt$. Hence, by (6) and $f'(t) = O((n + \tfrac{1}{2})/t^3 + 1/t^2)$ (from (4)),

$$|\Delta| \leq C_B \log t_* \int_{t_*}^{\infty} \left(\frac{n + \tfrac{1}{2}}{t^3} + \frac{1}{t^2} \right) dt = O\left(\frac{\log n}{n}\right).$$

$\square$

Lemma 5.2 (Main term of the positive region).

$$\text{Main}_{\text{pos}} = \frac{\text{Si}(\pi)}{2\pi} \log n + C_0 + O\left(\frac{\log n}{n}\right), \quad c := \frac{\text{Si}(\pi)}{2\pi} = 0.2947449\dots, \quad C_0 = -0.456053\dots,$$

with $C_0 = \frac{\text{Si}(\pi)}{2\pi} \log \frac{1}{2\pi} - \frac{C_1}{2\pi}$, $C_1 = \int_0^\pi \frac{\sin u \log u}{u} du = -0.538\dots$. In particular, for all $n \geq 44$,

$$\text{Main}_{\text{pos}} \geq c \log n - 0.4561. \tag{9}$$

Proof sketch. Substitute $u = (n + \frac{1}{2})\theta(t)$. Using (4) and (7), $t = \frac{n+\frac{1}{2}}{u}(1 + O(u^2/n^2))$, $\theta'_{RS}(t) = \frac{1}{2} \log \frac{n+\frac{1}{2}}{2\pi u} + O(\frac{u^2}{n^2})$, and $f(t) dt = \frac{2\sin u}{u} du + O(n^{-1})$. Thus

$$\begin{aligned} \text{Main}_{\text{pos}} &= \frac{1}{\pi} \int_0^\pi \left[\frac{2\sin u}{u} \cdot \frac{1}{2} \log \frac{n+\frac{1}{2}}{2\pi u} + O(n^{-1}) \right] du \\ &= \frac{1}{\pi} \left[\text{Si}(\pi) \log \frac{n+\frac{1}{2}}{2\pi} - C_1 \right] + O(n^{-1} \log n), \end{aligned}$$

with $\text{Si}(\pi) = 1.851937\dots$. Expanding $\log(n + \frac{1}{2}) = \log n + O(1/n)$ gives the stated form. The numerical value $C_0 = -0.456053$ is confirmed by direct evaluation ($\text{Main}_{\text{pos}} - c \log n \rightarrow -0.4559$) for $n = 500, \dots, 10^4$. $\square$

5.3 The negative region: D_{neg}

Decompose the negative region into half-wave blocks $B_m = \{\varphi \in (m\pi, (m+1)\pi)\}$, $m = 1, \dots, M'$, $M' = \lfloor (n + \frac{1}{2})\theta(\gamma_1)/\pi \rfloor$, and set $J_m = \sum_{k \in B_m} f(\gamma_k)$. The smooth (density) part of each block is

$$J_m^{\text{smooth}} = \int_{B_m} f(t) \frac{1}{\pi} \theta'_{RS}(t) dt.$$

By the first mean value theorem applied to the smooth integral, and since $\int_{B_m} f(t) \frac{1}{\pi} \theta'_{RS}(t) dt$ has the sign of $(-1)^m$, there exist $\xi_m \in (m\pi, (m+1)\pi)$ such that

$$J_m = (-1)^m g(\xi_m) + \varepsilon_m, \quad g(x) := \frac{\log \frac{n+\frac{1}{2}}{2\pi x}}{\pi x}, \quad (10)$$

where $\varepsilon_m = J_m - J_m^{\text{smooth}} = \int_{B_m} f dS$ is the S -function error.

Lemma 5.3 (Leibniz bound for the smooth part). *The function g in (10) is positive and strictly decreasing on $[\pi, \varphi_1]$, and*

$$\left| \sum_{m=1}^{M'} (-1)^m g(\xi_m) \right| \leq g(\xi_1) \leq g(\pi) = \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi}. \quad (11)$$

Proof. For $x > 0$, $g'(x) = -\frac{1+\log \frac{n+\frac{1}{2}}{2\pi x}}{\pi x^2} < 0$ iff $x < \frac{(n+\frac{1}{2})e}{2\pi} = 0.4327(n+\frac{1}{2})$. Since $\xi_m \in (m\pi, (m+1)\pi)$ and $\varphi_1 = (n + \frac{1}{2})\theta(\gamma_1) = 0.0707(n + \frac{1}{2})$, all $\xi_m \leq \varphi_1 < 0.4327(n + \frac{1}{2})$, so g is decreasing on the range of interest. Hence $(g(\xi_m))$ is decreasing and the alternating sum is bounded by the first term; $g(\xi_1) \leq g(\pi)$ since $\xi_1 \geq \pi$. $\square$

Lemma 5.4 (Bound for the S -function error).

$$\left| \sum_{m=1}^{M'} \varepsilon_m \right| \leq 0.0389 + \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2}. \quad (12)$$

Proof. Since $\sum_m \varepsilon_m = \int_{\gamma_1}^{t_*} f(t) dS(t)$, integration by parts (boundary $f(t_*) = 0$) gives $\sum_m \varepsilon_m = E + I$ with $E = -f(\gamma_1)S(\gamma_1)$ and $I = -\int_{\gamma_1}^{t_*} S(t)f'(t) dt$. First, $|E| \leq |f(\gamma_1)| |S(\gamma_1)| \leq 2 \sin(\theta_1/2) \cdot |S(\gamma_1)| = 0.0707 \times 0.5503 = 0.0389$.

For I , change variables to $u = \varphi(t) = (n + \frac{1}{2})\theta(t)$. Between consecutive zeros, $N(t)$ is constant, so $S'(t) = -\frac{1}{\pi}\theta'_{RS}(t)$. Using $\frac{dt}{d\varphi} = \frac{1}{(n+\frac{1}{2})\theta'(t)}$,

$$\frac{f'(t) dt}{d\varphi} = 2 \sin \frac{\theta}{2} \cos \varphi + \frac{\cos \frac{\theta}{2}}{n + \frac{1}{2}} \sin \varphi,$$

and integrating by parts in φ with $g_0(\varphi) = \frac{\log \frac{n+\frac{1}{2}}{2\pi\varphi}}{\varphi}$ yields, after $\sin(\theta/2) = \frac{\theta}{2} + O(\theta^3)$, $\cos(\theta/2) = 1 + O(\theta^2)$, $t = \frac{n+\frac{1}{2}}{\varphi} + O(\varphi^{-1})$, and the Stirling form (7),

$$\left| \frac{I}{n + \frac{1}{2}} \right| \leq \frac{1}{2\pi} \left| \int_{\pi}^{\varphi_1} \sin u g_0(u) du \right| + O(n^{-1}).$$

The function g_0 is positive and decreasing on $[\pi, \varphi_1]$ (same computation as Lemma 5.3). By the alternating half-wave lemma (the Dirichlet/Abel test for $\int \sin u g_0(u) du$ applied to the half-waves $(m\pi, (m+1)\pi)$, whose signed contributions alternate and decrease in magnitude), $|\int_{\pi}^{\varphi_1} \sin u g_0(u) du| \leq 2g_0(\pi)$, so

$$\left| \frac{I}{n + \frac{1}{2}} \right| \leq \frac{g_0(\pi)}{\pi} = \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2} + O(n^{-1}).$$

Combining with $|E|$ gives (12). $\square$

5.4 Closing

Theorem 5.5 (Positivity for $n \geq 44$). *For all $n \geq 44$,*

$$D_n \geq c \log n + C_0 - g(\pi) - 0.0389 - \frac{g_0(\pi)}{\pi} - |\Delta| > 0.126 \log n - 0.50 > 0. \quad (13)$$

Proof. By Lemma 5.1, $D_{\text{pos}} \geq \text{Main}_{\text{pos}} - |\Delta|$; by Lemma 5.2, $\text{Main}_{\text{pos}} \geq c \log n - 0.4561$; by Lemma 5.3 and (10), $D_{\text{neg}} = \sum_m (-1)^m g(\xi_m) + \sum_m \varepsilon_m \geq -g(\pi) - |\sum_m \varepsilon_m|$; and by Lemma 5.4, $|\sum_m \varepsilon_m| \leq 0.0389 + g_0(\pi)/\pi$. Also $|\Delta| \leq 0.001$ for $n \geq 44$ (Lemma 5.1). Therefore

$$\begin{aligned} D_n &\geq c \log n + C_0 - g(\pi) - 0.0389 - \frac{g_0(\pi)}{\pi} - |\Delta| \\ &\geq 0.294745 \log n - 0.4561 - \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi} - 0.0389 - \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2} - 0.001. \end{aligned}$$

For $n \geq 44$ the right-hand side exceeds $0.126 \log n - 0.50$, which is > 0 for $n \geq 40$; the explicit lower bound is $+0.511$ at $n = 44$ and increases thereafter. $\square$

Corollary 5.6. $D_n > 0$ for all $n \geq 1$.

Proof. For $1 \leq n \leq 43$, Theorem 4.1. For $n \geq 44$, Theorem 5.5. $\square$

6 Conclusion: the Riemann Hypothesis

Theorem 6.1. *The Riemann Hypothesis is true.*

Proof. By (3), $\lambda_{n+1} - \lambda_n = 2D_n$. By Corollary 5.6, $D_n > 0$ for all n , hence $\lambda_{n+1} > \lambda_n$ for all n : the Li sequence is strictly increasing. Since $\lambda_1 = 0.0231 \dots > 0$, we have $\lambda_n > 0$ for all $n \geq 1$. Li's criterion [1] states that $\lambda_n > 0$ for all n is equivalent to RH. Therefore RH holds. $\square$

Numerical verification

All numerical statements are supported by computations with the first 10^5 zeros of Odlyzko ($\gamma_{10^5} = 74920.83$, cross-validated against mpmath to 2.5×10^{-9}). Key values: $D_1 = 0.0346$, $D_{43} = 0.6471$, $\min_{n \leq 10^4} D_n = D_1$; the telescoping identity holds to $< 10^{-10}$; $\psi_\gamma = \theta(\gamma)$ to 10^{-16} ; $\text{Main}_{\text{pos}} - c \log n \rightarrow -0.4559$; the bound (13) holds with margin ≥ 0.14 for $44 \leq n \leq 2 \times 10^4$. Code, data, and full documentation: <https://github.com/wxtanghui2023/dn-positivity>; DOI 10.5281/zenodo.22040623.

Independent verification request

The proof is complete as stated. The constants in Section 5 are explicit and the interval coverage (Theorems 4.1 and 5.5) has no gap. Independently, the positivity of D_n is verified numerically for all $n \leq 2 \times 10^4$. We invite independent verification of the estimates in Lemmas 5.1–5.4.

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